

Note 4.1 Definition and Examples

Definition

A mapping L from a vector space V into a vector space W is said to be a linear transformation if

$$L(\alpha \mathbf{v}_1 + \beta \mathbf{v}_2) = \alpha L(\mathbf{v}_1) + \beta L(\mathbf{v}_2)$$

for all $\mathbf{v}_1, \mathbf{v}_2 \in V$ and for all scalars α, β .

It is the same as

$$L(\mathbf{v}_1 + \mathbf{v}_2) = L(\mathbf{v}_1) + L(\mathbf{v}_2) \quad (\alpha = \beta = 1)$$

$$L(\alpha \mathbf{v}) = \alpha L(\mathbf{v}) \quad (\mathbf{v} = \mathbf{v}_1, \beta = 0)$$

Notation

A mapping L from a vector space V into a vector space W will be denoted by

$$L : V \rightarrow W$$

If the two vector spaces are the same, we say $L : V \rightarrow V$ is a linear operator on V .

Linear Operators on $\mathbb{R}^2$

The stretch, projection, rotation and reflecting are all the linear transformation.

Linear Transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$

In general, if we want to linearly transform the vector from $\mathbb{R}^n$ to $\mathbb{R}^m$, then we need to ensure a $m \times n$ matrix A such that

$$L_A(\mathbf{x}) = A\mathbf{x}$$

In fact, any linear transformation can be expressed by matrix multiplication. It will be showed next section.

Linear Transformation from V to W

If L is a linear transformation mapping a vector space V into a vector space W , then

- $L(\mathbf{0}_V) = \mathbf{0}_W$
- $L(\alpha_1 \mathbf{v}_1 + \cdots + \alpha_n \mathbf{v}_n) = \alpha_1 L(\mathbf{v}_1) + \cdots + \alpha_n L(\mathbf{v}_n)$
- $L(-\mathbf{v}) = -L(\mathbf{v})$ for all $\mathbf{v} \in V$

The Image and Kernel

Definition of Kernel

Let $L : V \rightarrow W$ be a linear transformation, then the **kernel** of L , denoted by $\ker(L)$, is **the set of all vectors in V whose mapping in W is zero vectors**. That is,

$$\ker(L) = \{\mathbf{v} \in V \mid L(\mathbf{v}) = \mathbf{0}_W\}$$

Definition of Image

Let $L : V \rightarrow W$ be a linear transformation and let S be a subspace of V . The **image** of S , denoted by $L(S)$, is **the set of vectors mapping into W** . That is,

$$L(S) = \{\mathbf{w} \in W \mid \mathbf{w} = L(\mathbf{v}) \text{ for some } \mathbf{v} \in S\}$$

The image of the entire vector space, $L(V)$, is called the **range** of L .

Theorem 4.1.1

Let $L : V \rightarrow W$ be a linear transformation and let S be a subspace of V , then

- $\ker(L)$ is a subspace of V .
- $L(S)$ is a subspace of W .